

The Complete Time-Dependent Covariance Matrix of the Stochastic Damped Harmonic Oscillator

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Abstract

We present a comprehensive and rigorous derivation of the complete time-dependent covariance matrix for the stochastic damped harmonic oscillator governed by the Langevin equation $x''(t) + ax'(t) + bx(t) = \sigma\eta(t)$, where $\eta(t)$ is Gaussian white noise. While previous treatments have focused primarily on the position marginal or stationary statistics, we derive explicit closed-form expressions for the full 2×2 covariance structure, including the position-velocity cross-correlation, for all damping regimes: overdamped, critically damped, and underdamped. Our results reveal that the position and velocity are correlated at finite times, with the cross-correlation taking a remarkably simple form $\Sigma_{12}(t) = \frac{\sigma^2}{2}G(t)^2$, where $G(t)$ is the Green's function of the system. The cross-correlation vanishes only in the stationary limit. We verify that all expressions reduce correctly to the known stationary covariance matrix obtained from the Lyapunov equation, confirming the equipartition theorem. The formulas presented here provide a complete statistical characterization of the process, with applications in stochastic control, statistical mechanics, and data assimilation.

1 Introduction

The stochastic differential equation

$$x''(t) + ax'(t) + bx(t) = \sigma\eta(t), \quad (1)$$

where $\eta(t)$ is zero-mean Gaussian white noise with autocorrelation $\langle \eta(t)\eta(s) \rangle = \delta(t-s)$, serves as a fundamental model across numerous scientific disciplines. It describes the Brownian motion of a particle in a harmonic potential [1, 2], electrical fluctuations in RLC circuits [3], and mean-reverting processes in financial mathematics [4].

The complete statistical characterization of this system requires knowledge of the joint probability distribution of position and velocity. Since the process is Gaussian for deterministic initial conditions, this distribution is fully determined by its mean vector and covariance matrix:

$$\Sigma(t) = \begin{pmatrix} \Sigma_{11}(t) & \Sigma_{12}(t) \\ \Sigma_{21}(t) & \Sigma_{22}(t) \end{pmatrix} = \begin{pmatrix} \text{Var}[x(t)] & \text{Cov}[x(t), v(t)] \\ \text{Cov}[v(t), x(t)] & \text{Var}[v(t)] \end{pmatrix}. \quad (2)$$

Despite the extensive literature on this system, many treatments focus exclusively on the marginal distribution of the position or on the stationary statistics. The complete

time-dependent joint distribution, particularly the position-velocity cross-correlation $\Sigma_{12}(t)$, has not been systematically presented in a unified framework covering all damping regimes.

This paper provides a complete and rigorous derivation of the full covariance matrix for all damping regimes. We show that the cross-covariance takes an exceptionally simple form in terms of the Green's function, and we provide explicit closed-form expressions that are manifestly positive-definite.

2 Mathematical Formulation

2.1 State-Space Representation

We rewrite (1) as a first-order system in the Itô sense:

$$d\mathbf{X}(t) = A\mathbf{X}(t) dt + B dW(t), \quad (3)$$

where

$$\mathbf{X}(t) = \begin{pmatrix} x(t) \\ v(t) \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \sigma \end{pmatrix}. \quad (4)$$

Here $W(t)$ is a standard Wiener process. For additive noise, the Itô and Stratonovich interpretations coincide, so no correction term is needed.

The solution of (3) is

$$\mathbf{X}(t) = e^{At}\mathbf{X}(0) + \int_0^t e^{A(t-s)}B dW(s). \quad (5)$$

For deterministic initial conditions $\mathbf{X}(0)$, the covariance matrix evolves as

$$\Sigma(t) = \int_0^t e^{A\tau}BB^Te^{A^T\tau} d\tau. \quad (6)$$

2.2 The Green's Function

The Green's function $G(t)$ satisfies:

$$G'' + aG' + bG = \delta(t), \quad G(0) = 0, \quad \dot{G}(0^+) = 1, \quad G(t) = 0 \text{ for } t < 0. \quad (7)$$

Here $\dot{G}(0^+)$ denotes the right-hand derivative at zero, representing the jump condition imposed by the delta function. The matrix exponential e^{At} can be expressed in terms of $G(t)$ and its derivatives for $t > 0$:

$$e^{At} = \begin{pmatrix} \dot{G}(t) + aG(t) & G(t) \\ -bG(t) & \dot{G}(t) \end{pmatrix}. \quad (8)$$

This structure will prove essential for deriving the covariance matrix elements. Only the (1, 2) and (2, 2) entries of e^{At} are needed for the covariance calculations.

3 General Covariance Matrix Elements

From (6) and the form $BB^T = \begin{pmatrix} 0 & 0 \\ 0 & \sigma^2 \end{pmatrix}$, we obtain:

$$\Sigma_{11}(t) = \sigma^2 \int_0^t [e^{A\tau}]_{12}^2 d\tau = \sigma^2 \int_0^t G(\tau)^2 d\tau, \quad (9)$$

$$\Sigma_{12}(t) = \sigma^2 \int_0^t [e^{A\tau}]_{12} [e^{A\tau}]_{22} d\tau = \sigma^2 \int_0^t G(\tau) \dot{G}(\tau) d\tau, \quad (10)$$

$$\Sigma_{22}(t) = \sigma^2 \int_0^t [e^{A\tau}]_{22}^2 d\tau = \sigma^2 \int_0^t \dot{G}(\tau)^2 d\tau. \quad (11)$$

A key observation is that since

$$\frac{d}{d\tau} \left[\frac{1}{2} G(\tau)^2 \right] = G(\tau) \dot{G}(\tau),$$

we have the remarkably simple result:

$$\boxed{\Sigma_{12}(t) = \frac{\sigma^2}{2} G(t)^2.} \quad (12)$$

This holds for all damping regimes. The cross-covariance between position and velocity is simply proportional to the square of the Green's function.

4 Overdamped Regime ($a^2 > 4b$)

4.1 The Green's Function

Let $r_1 > r_2$ denote the two distinct negative real roots of the characteristic equation $r^2 + ar + b = 0$. Since $a > 0$ and $b > 0$, we have $r_1, r_2 < 0$. The Green's function is

$$G(t) = \frac{e^{r_1 t} - e^{r_2 t}}{r_1 - r_2}, \quad t \geq 0. \quad (13)$$

4.2 Position Variance

From (9):

$$\begin{aligned} \Sigma_{11}(t) &= \sigma^2 \int_0^t G(s)^2 ds \\ &= \frac{\sigma^2}{(r_1 - r_2)^2} \left[\int_0^t e^{2r_1 s} ds - 2 \int_0^t e^{(r_1 + r_2)s} ds + \int_0^t e^{2r_2 s} ds \right]. \end{aligned} \quad (14)$$

Evaluating the integrals yields:

$$\boxed{\Sigma_{11}(t) = \frac{\sigma^2}{(r_1 - r_2)^2} \left[\frac{e^{2r_1 t} - 1}{2r_1} - \frac{2(e^{(r_1 + r_2)t} - 1)}{r_1 + r_2} + \frac{e^{2r_2 t} - 1}{2r_2} \right].} \quad (15)$$

This expression is manifestly positive for all $t > 0$, since it equals $\sigma^2 \int_0^t G(s)^2 ds$.

4.3 Position-Velocity Cross-Covariance

Using (12):

$$\Sigma_{12}(t) = \frac{\sigma^2}{2} \left(\frac{e^{r_1 t} - e^{r_2 t}}{r_1 - r_2} \right)^2. \quad (16)$$

Equivalently, in expanded form:

$$\Sigma_{12}(t) = \frac{\sigma^2}{2(r_1 - r_2)^2} [e^{2r_1 t} - 2e^{(r_1+r_2)t} + e^{2r_2 t}]. \quad (17)$$

4.4 Velocity Variance

Since

$$\dot{G}(t) = \frac{r_1 e^{r_1 t} - r_2 e^{r_2 t}}{r_1 - r_2}, \quad (18)$$

we have

$$\begin{aligned} \Sigma_{22}(t) &= \sigma^2 \int_0^t \dot{G}(s)^2 ds \\ &= \frac{\sigma^2}{(r_1 - r_2)^2} \left[\frac{r_1^2 (e^{2r_1 t} - 1)}{2r_1} - \frac{2r_1 r_2 (e^{(r_1+r_2)t} - 1)}{r_1 + r_2} + \frac{r_2^2 (e^{2r_2 t} - 1)}{2r_2} \right]. \end{aligned} \quad (19)$$

Simplifying:

$$\Sigma_{22}(t) = \frac{\sigma^2}{(r_1 - r_2)^2} \left[\frac{r_1 (e^{2r_1 t} - 1)}{2} - \frac{2r_1 r_2 (e^{(r_1+r_2)t} - 1)}{r_1 + r_2} + \frac{r_2 (e^{2r_2 t} - 1)}{2} \right]. \quad (20)$$

The positivity of this expression follows from the integral representation $\Sigma_{22}(t) = \sigma^2 \int_0^t \dot{G}(s)^2 ds \geq 0$.

5 Critically Damped Regime ($a^2 = 4b$)

5.1 The Green's Function

Let $r = -a/2 < 0$ be the repeated root. The Green's function is

$$G(t) = t e^{rt}, \quad t \geq 0. \quad (21)$$

5.2 Position Variance

$$\begin{aligned} \Sigma_{11}(t) &= \sigma^2 \int_0^t s^2 e^{2rs} ds \\ &= \sigma^2 \left[e^{2rs} \left(\frac{s^2}{2r} - \frac{s}{2r^2} + \frac{1}{4r^3} \right) \right]_0^t. \end{aligned} \quad (22)$$

Thus:

$$\Sigma_{11}(t) = \frac{\sigma^2}{4|r|^3} [1 - e^{2rt} (1 - 2rt + 2r^2 t^2)]. \quad (23)$$

With $\lambda = -r > 0$:

$$\Sigma_{11}(t) = \frac{\sigma^2}{4\lambda^3} [1 - e^{-2\lambda t} (1 + 2\lambda t + 2\lambda^2 t^2)] > 0. \quad (24)$$

5.3 Position-Velocity Cross-Covariance

Using (12):

$$\Sigma_{12}(t) = \frac{\sigma^2 t^2}{2} e^{2rt}. \quad (25)$$

5.4 Velocity Variance

Since

$$\dot{G}(t) = (1 + rt)e^{rt}, \quad (26)$$

we have

$$\begin{aligned} \Sigma_{22}(t) &= \sigma^2 \int_0^t (1 + rs)^2 e^{2rs} ds \\ &= \sigma^2 \left[e^{2rt} \left(\frac{1}{4r} + \frac{t}{2} + \frac{rt^2}{2} \right) - \frac{1}{4r} \right]. \end{aligned} \quad (27)$$

Thus:

$$\Sigma_{22}(t) = \sigma^2 \left[e^{2rt} \left(\frac{1}{4r} + \frac{t}{2} + \frac{rt^2}{2} \right) - \frac{1}{4r} \right]. \quad (28)$$

6 Underdamped Regime ($a^2 < 4b$)

6.1 The Green's Function

Define

$$\alpha = \frac{a}{2} > 0, \quad \omega = \sqrt{b - \frac{a^2}{4}} > 0, \quad b = \alpha^2 + \omega^2. \quad (29)$$

The Green's function is

$$G(t) = \frac{e^{-\alpha t}}{\omega} \sin(\omega t), \quad t \geq 0. \quad (30)$$

6.2 Position Variance

$$\begin{aligned} \Sigma_{11}(t) &= \frac{\sigma^2}{\omega^2} \int_0^t e^{-2\alpha s} \sin^2(\omega s) ds \\ &= \frac{\sigma^2}{2\omega^2} \left[\int_0^t e^{-2\alpha s} ds - \int_0^t e^{-2\alpha s} \cos(2\omega s) ds \right]. \end{aligned} \quad (31)$$

Using the standard integrals:

$$\int_0^t e^{-2\alpha s} ds = \frac{1 - e^{-2\alpha t}}{2\alpha}, \quad (32)$$

$$\int_0^t e^{-2\alpha s} \cos(2\omega s) ds = \frac{\alpha(1 - e^{-2\alpha t} \cos 2\omega t) + \omega e^{-2\alpha t} \sin 2\omega t}{2(\alpha^2 + \omega^2)}. \quad (33)$$

Therefore:

$$\Sigma_{11}(t) = \frac{\sigma^2}{2\omega^2} \left[\frac{1 - e^{-2\alpha t}}{2\alpha} - \frac{\alpha(1 - e^{-2\alpha t} \cos 2\omega t) + \omega e^{-2\alpha t} \sin 2\omega t}{2(\alpha^2 + \omega^2)} \right]. \quad (34)$$

6.3 Position-Velocity Cross-Covariance

Using (12):

$$\Sigma_{12}(t) = \frac{\sigma^2 e^{-2\alpha t}}{2\omega^2} \sin^2(\omega t). \quad (35)$$

6.4 Velocity Variance

Since

$$\dot{G}(t) = e^{-\alpha t} \left[\cos(\omega t) - \frac{\alpha}{\omega} \sin(\omega t) \right], \quad (36)$$

we have

$$\Sigma_{22}(t) = \sigma^2 \int_0^t \dot{G}(s)^2 ds. \quad (37)$$

After evaluating the integral (see Appendix A for details):

$$\Sigma_{22}(t) = \frac{\sigma^2}{4\alpha} - \frac{\sigma^2 e^{-2\alpha t}}{4\omega^2} \left[\frac{b}{\alpha} - \alpha \cos(2\omega t) - \omega \sin(2\omega t) \right]. \quad (38)$$

7 Stationary Limits and Verification

7.1 Stationary Covariance Matrix

Taking the limit $t \rightarrow \infty$ in all regimes yields:

$$\Sigma_{\text{stat}} = \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix}. \quad (39)$$

7.2 Verification of Equipartition

The stationary variances satisfy:

$$\text{Var}[x]_{\text{stat}} = \frac{\sigma^2}{2ab} = \frac{k_B T}{b}, \quad (40)$$

$$\text{Var}[v]_{\text{stat}} = \frac{\sigma^2}{2a} = \frac{k_B T}{m}, \quad (41)$$

where we identify $\sigma^2/(2a) = k_B T/m$. This confirms the equipartition theorem: each degree of freedom has energy $\frac{1}{2}k_B T$.

7.3 Lyapunov Equation Verification

The stationary covariance matrix satisfies the Lyapunov equation:

$$A\Sigma_{\text{stat}} + \Sigma_{\text{stat}}A^T + BB^T = 0. \quad (42)$$

Verification:

$$\begin{aligned} A\Sigma_{\text{stat}} + \Sigma_{\text{stat}}A^T &= \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix} \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix} + \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix} \begin{pmatrix} 0 & -b \\ 1 & -a \end{pmatrix} \\ &= \begin{pmatrix} 0 & \frac{\sigma^2}{2a} \\ -\frac{\sigma^2}{2a} & -\frac{\sigma^2}{2} \end{pmatrix} + \begin{pmatrix} 0 & -\frac{\sigma^2}{2a} \\ \frac{\sigma^2}{2a} & -\frac{\sigma^2}{2} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -\sigma^2 \end{pmatrix}. \end{aligned} \quad (43)$$

Adding $BB^T = \begin{pmatrix} 0 & 0 \\ 0 & \sigma^2 \end{pmatrix}$ gives zero. $\checkmark$

8 The Correlation Coefficient

The correlation coefficient between position and velocity is:

$$\rho(t) = \frac{\Sigma_{12}(t)}{\sqrt{\Sigma_{11}(t)\Sigma_{22}(t)}}. \quad (44)$$

Using (12):

$$\rho(t) = \frac{\frac{\sigma^2}{2}G(t)^2}{\sqrt{\Sigma_{11}(t)\Sigma_{22}(t)}}. \quad (45)$$

Properties:

- (i) **Initial condition:** $\rho(0) = 0$ (deterministic initial conditions)
- (ii) **Stationary limit:** $\rho(\infty) = 0$ (position and velocity are uncorrelated at stationarity)
- (iii) **Sign:** In the overdamped and critically damped cases, $\rho(t) > 0$ for all $t > 0$, indicating positive correlation between position and velocity
- (iv) **Underdamped:** $\rho(t)$ oscillates with frequency 2ω and decays with rate 2α

9 Special Cases and Asymptotics

9.1 Undamped Limit ($a \rightarrow 0$)

As $a \rightarrow 0$, the system becomes a stochastic undamped oscillator with $\omega = \sqrt{b}$:

$$G(t) = \frac{1}{\sqrt{b}} \sin(\sqrt{b}t). \quad (46)$$

The covariance elements become:

$$\Sigma_{11}(t) = \frac{\sigma^2}{2b\sqrt{b}} \left[t\sqrt{b} - \frac{1}{2} \sin(2\sqrt{b}t) \right], \quad (47)$$

$$\Sigma_{12}(t) = \frac{\sigma^2}{2b} \sin^2(\sqrt{b}t), \quad (48)$$

$$\Sigma_{22}(t) = \frac{\sigma^2}{2\sqrt{b}} \left[t\sqrt{b} + \frac{1}{2} \sin(2\sqrt{b}t) \right]. \quad (49)$$

Note that in the limit $a \rightarrow 0$, the stationary covariance matrix does not exist, as the system has no damping and energy grows linearly with time.

9.2 Free Particle Limit ($b \rightarrow 0$)

As $b \rightarrow 0$, the oscillator becomes a free Brownian particle. The roots are $r_1 \rightarrow -b/a$ and $r_2 \rightarrow -a$:

$$\Sigma_{11}(t) = \frac{\sigma^2}{a^3} \left[at - 2(1 - e^{-at}) + \frac{1}{2}(1 - e^{-2at}) \right], \quad (50)$$

$$\Sigma_{12}(t) = \frac{\sigma^2}{2a^2} (1 - e^{-at})^2, \quad (51)$$

$$\Sigma_{22}(t) = \frac{\sigma^2}{2a} (1 - e^{-2at}). \quad (52)$$

9.3 Heavy Damping Limit ($a \gg \sqrt{b}$)

For strong damping:

$$\Sigma_{11}(t) \approx \frac{\sigma^2}{2ab} (1 - e^{-2bt/a}), \quad (53)$$

$$\Sigma_{12}(t) \approx 0, \quad (54)$$

$$\Sigma_{22}(t) \approx \frac{\sigma^2}{2a} (1 - e^{-2at}). \quad (55)$$

9.4 Weak Damping Limit ($a \ll \sqrt{b}$)

For weak damping:

$$\Sigma_{11}(t) \approx \frac{\sigma^2}{2ab} \left[1 - e^{-at} \left(\cos^2 \omega t + \frac{a}{2\omega} \sin 2\omega t \right) \right], \quad (56)$$

$$\Sigma_{12}(t) \approx \frac{\sigma^2}{2b} e^{-at} \sin^2 \omega t, \quad (57)$$

$$\Sigma_{22}(t) \approx \frac{\sigma^2}{2a} \left[1 - e^{-at} \left(\cos^2 \omega t + \frac{a}{2\omega} \sin 2\omega t \right) \right]. \quad (58)$$

10 Physical Interpretation

10.1 The Green's Function as a Correlation Generator

The fact that $\Sigma_{12}(t) = \frac{\sigma^2}{2} G(t)^2$ has a beautiful physical interpretation: the cross-covariance between position and velocity is determined entirely by the system's impulse response. This reflects the fact that the stochastic forcing at time s affects both position and velocity at time t through the Green's function.

10.2 Positive Correlation in Overdamped and Critically Damped Cases

For overdamped and critically damped systems, $G(t) > 0$ for all $t > 0$, so $\Sigma_{12}(t) > 0$. This means:

- A positive displacement $x(t) > 0$ tends to be associated with a positive velocity $v(t) > 0$

- The particle moves away from equilibrium in the direction of its displacement

This is physically intuitive: a particle displaced from equilibrium has momentum in the direction of displacement.

10.3 Oscillating Correlation in Underdamped Case

In the underdamped case, $G(t) = \frac{e^{-\alpha t}}{\omega} \sin(\omega t)$, so:

$$\Sigma_{12}(t) = \frac{\sigma^2 e^{-2\alpha t}}{2\omega^2} \sin^2(\omega t). \quad (59)$$

The cross-covariance oscillates with frequency 2ω and decays with rate 2α . This reflects the oscillatory nature of the underdamped dynamics. Although the Green's function changes sign, its square is always non-negative, so the correlation is always positive but oscillates in magnitude.

11 Conclusion

We have derived explicit closed-form expressions for the complete time-dependent covariance matrix of the stochastic damped harmonic oscillator in all damping regimes. The key results are:

1. **The simple form of the cross-covariance:** $\Sigma_{12}(t) = \frac{\sigma^2}{2} G(t)^2$, valid for all damping regimes.
2. **Complete characterization:** The formulas for $\Sigma_{11}(t)$, $\Sigma_{12}(t)$, and $\Sigma_{22}(t)$ fully determine the joint distribution of position and velocity at all times.
3. **Stationary limits:** All expressions converge to the known stationary covariance matrix, confirming the equipartition theorem:

$$\Sigma_{\text{stat}} = \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix}.$$

The results presented here have immediate applications in:

- **Stochastic control:** optimal state estimation and Kalman filtering
- **Statistical mechanics:** computation of time-dependent response functions
- **Data assimilation:** combining model predictions with observations
- **Signal processing:** characterization of noise in physical systems

A Detailed Derivation of Underdamped Velocity Variance

From (11) and (36):

$$\dot{G}(t)^2 = e^{-2\alpha t} \left[\cos(\omega t) - \frac{\alpha}{\omega} \sin(\omega t) \right]^2. \quad (60)$$

Expanding:

$$\begin{aligned} \dot{G}(t)^2 &= e^{-2\alpha t} \left[\cos^2(\omega t) - \frac{2\alpha}{\omega} \sin(\omega t) \cos(\omega t) + \frac{\alpha^2}{\omega^2} \sin^2(\omega t) \right] \\ &= e^{-2\alpha t} \left[\left(\frac{1}{2} + \frac{\alpha^2}{2\omega^2} \right) + \left(\frac{1}{2} - \frac{\alpha^2}{2\omega^2} \right) \cos 2\omega t - \frac{\alpha}{\omega} \sin 2\omega t \right] \\ &= e^{-2\alpha t} \left[\frac{b}{2\omega^2} + \frac{\omega^2 - \alpha^2}{2\omega^2} \cos 2\omega t - \frac{\alpha}{\omega} \sin 2\omega t \right]. \end{aligned} \quad (61)$$

Integrating from 0 to t :

$$\Sigma_{22}(t) = \sigma^2 \left[\frac{b}{2\omega^2} \int_0^t e^{-2\alpha s} ds + \frac{\omega^2 - \alpha^2}{2\omega^2} \int_0^t e^{-2\alpha s} \cos(2\omega s) ds - \frac{\alpha}{\omega} \int_0^t e^{-2\alpha s} \sin(2\omega s) ds \right]. \quad (62)$$

Using the standard integrals:

$$\int_0^t e^{-2\alpha s} ds = \frac{1 - e^{-2\alpha t}}{2\alpha}, \quad (63)$$

$$\int_0^t e^{-2\alpha s} \cos(2\omega s) ds = \frac{\alpha(1 - e^{-2\alpha t} \cos 2\omega t) - \omega e^{-2\alpha t} \sin 2\omega t}{2(\alpha^2 + \omega^2)}, \quad (64)$$

$$\int_0^t e^{-2\alpha s} \sin(2\omega s) ds = \frac{\omega(1 - e^{-2\alpha t} \cos 2\omega t) + \alpha e^{-2\alpha t} \sin 2\omega t}{2(\alpha^2 + \omega^2)}. \quad (65)$$

Substituting and simplifying yields (38). $\square$

B Verification of the Matrix Exponential

The matrix exponential (8) satisfies the necessary properties for $t > 0$:

1. **Initial condition:**

$$\lim_{t \rightarrow 0^+} e^{At} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

2. **Derivative at $t = 0^+$:**

$$\left. \frac{d}{dt} e^{At} \right|_{t=0^+} = A$$

3. **Differential equation:**

$$\frac{d}{dt} e^{At} = A e^{At}, \quad t > 0$$

These properties are satisfied by (8) due to the Green's function equations $G(0) = 0$, $\dot{G}(0^+) = 1$, and $\ddot{G} + a\dot{G} + bG = 0$ for $t > 0$.

C Summary of Covariance Matrix Elements

We collect the closed-form expressions for all three damping regimes.

Overdamped Regime ($a^2 > 4b$)

Roots: $r_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2} < 0$, $r_1 > r_2$.

$$\Sigma_{11}(t) = \frac{\sigma^2}{(r_1 - r_2)^2} \left[\frac{e^{2r_1 t} - 1}{2r_1} - \frac{2(e^{(r_1+r_2)t} - 1)}{r_1 + r_2} + \frac{e^{2r_2 t} - 1}{2r_2} \right], \quad (66)$$

$$\Sigma_{12}(t) = \frac{\sigma^2}{2} \left(\frac{e^{r_1 t} - e^{r_2 t}}{r_1 - r_2} \right)^2, \quad (67)$$

$$\Sigma_{22}(t) = \frac{\sigma^2}{(r_1 - r_2)^2} \left[\frac{r_1(e^{2r_1 t} - 1)}{2} - \frac{2r_1 r_2(e^{(r_1+r_2)t} - 1)}{r_1 + r_2} + \frac{r_2(e^{2r_2 t} - 1)}{2} \right]. \quad (68)$$

Critically Damped Regime ($a^2 = 4b$)

Repeated root: $r = -a/2 < 0$, $\lambda = -r > 0$.

$$\Sigma_{11}(t) = \frac{\sigma^2}{4\lambda^3} \left[1 - e^{-2\lambda t} (1 + 2\lambda t + 2\lambda^2 t^2) \right], \quad (69)$$

$$\Sigma_{12}(t) = \frac{\sigma^2 t^2}{2} e^{-2\lambda t}, \quad (70)$$

$$\Sigma_{22}(t) = \sigma^2 \left[e^{-2\lambda t} \left(\frac{t}{2} - \frac{\lambda t^2}{2} - \frac{1}{4\lambda} \right) + \frac{1}{4\lambda} \right]. \quad (71)$$

Underdamped Regime ($a^2 < 4b$)

Parameters: $\alpha = a/2$, $\omega = \sqrt{b - \alpha^2}$.

$$\Sigma_{11}(t) = \frac{\sigma^2}{2\omega^2} \left[\frac{1 - e^{-2\alpha t}}{2\alpha} - \frac{\alpha(1 - e^{-2\alpha t} \cos 2\omega t) - \omega e^{-2\alpha t} \sin 2\omega t}{2(\alpha^2 + \omega^2)} \right], \quad (72)$$

$$\Sigma_{12}(t) = \frac{\sigma^2 e^{-2\alpha t}}{2\omega^2} \sin^2(\omega t), \quad (73)$$

$$\Sigma_{22}(t) = \frac{\sigma^2}{4\alpha} - \frac{\sigma^2 e^{-2\alpha t}}{4\omega^2} \left[\frac{b}{\alpha} - \alpha \cos(2\omega t) - \omega \sin(2\omega t) \right]. \quad (74)$$

Stationary Limits (all regimes)

$$\Sigma_{11}(\infty) = \frac{\sigma^2}{2ab}, \quad \Sigma_{12}(\infty) = 0, \quad \Sigma_{22}(\infty) = \frac{\sigma^2}{2a}.$$

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